

Computer Vision Assignment Use Cases with Datasets Department of Computer Science Gujarat University M.Sc (CS) Semester 6

Smart Surveillance System

Use CCTV video to detect motion using preprocessing, thresholding and morphology. Dataset: PETS Dataset, any CCTV footage.

Medical Image Analysis

Enhance MRI/X-ray images using histogram equalization and segmentation. Dataset: Brain MRI (Kaggle), Chest X-ray.

Document Processing

Convert scanned docs to binary using adaptive thresholding. Dataset: IAM Handwriting Dataset, scanned PDFs.

Autonomous Driving

Detect lanes using edge detection and smoothing. Dataset: KITTI dataset.

Noise Removal

Apply median and Gaussian filters. Dataset: Any image with added noise.

Thresholding

Compare simple, Otsu, adaptive thresholding. Dataset: Coins image (OpenCV).

K-Means Segmentation

Segment image into clusters. Dataset: Natural images.

Morphological Operations

Use erosion, dilation, opening, closing. Dataset: Binary images.

Video Processing

Extract and process frames. Dataset: Any MP4 video, UCF101.

Transformations

Apply rotation, scaling, translation. Dataset: Any image.